Comparative Analysis of Machine Learning Models for Customer Churn Prediction

William Arwan

Independent Researcher

Abstract

This paper presents a comparative evaluation of three machine learning architectures for predicting customer churn in a telecommunications dataset comprising 7,043 customers and 21 features. A baseline Random Forest classifier, a class-weighted Logistic Regression model, and a Multi-Layer Perceptron (MLP) neural network are trained and evaluated on a held-out test set of 1407 customers (374 churners). Results demonstrate that the class-weighted Logistic Regression achieves 79% churn recall compared to 48% for the baseline, representing a 32 percentage-point improvement. Feature importance analysis identifies contract type, tenure, and monthly charges as the primary churn drivers. The findings suggest that recall-optimized linear models offer superior practical value over accuracy-optimized ensemble methods for imbalanced churn prediction tasks.

Index Terms — customer churn, classification, class imbalance, logistic regression, random forest, multi-layer perceptron, ROC-AUC

I. Introduction

Customer churn prediction is a critical business intelligence task in the telecommunications industry, where acquiring a new customer costs five to seven times more than retaining an existing one [1]. Accurately identifying at-risk customers enables proactive retention strategies that can significantly reduce revenue loss.

The primary challenge in churn prediction arises from class imbalance: typically only 20-30% of customers churn in a given period. Standard classifiers optimized for overall accuracy tend to underpredict the minority class, producing models that appear performant but fail to identify the customers who matter most [2].

This study evaluates three distinct modeling strategies on the IBM Telco Customer Churn dataset: (a) a baseline Random Forest, (b) a class-weighted Logistic Regression, and (c) a hyperparameter-tuned MLP neural network. The objective is to maximize churn recall while maintaining acceptable precision.

II. Dataset and Preprocessing

The Telco Customer Churn dataset [3] contains 7,043 customer records with 21 features including demographics (gender, senior citizen status), account information (tenure, contract type, payment method), and service subscriptions (internet, phone, streaming). The binary target variable indicates whether the customer churned (27%) or not (73%).

Preprocessing consists of four steps: (1) removal of the non-predictive customerID column, (2) coercion of TotalCharges to numeric with missing value elimination (11 rows), (3) one-hot encoding of categorical variables with first-category dropping, and (4) standard scaling of numerical features. The data is split 80/20 for training and testing with a fixed random seed for reproducibility.

III. Methodology

A. Baseline Random Forest

A Random Forest classifier with 100 estimators is trained using default scikit-learn parameters. This model serves as the accuracy-optimized baseline and provides feature importance rankings via Gini impurity reduction.

B. Class-Weighted Logistic Regression

A Logistic Regression model with class_weight='balanced' is trained to penalize misclassification of the minority class proportionally to its inverse frequency. This approach adjusts the decision boundary to favor recall at the expense of precision.

C. Multi-Layer Perceptron Neural Network

An MLP classifier is optimized via RandomizedSearchCV over hidden layer sizes [(64,32), (128,64,32), (32,16)], activation functions [relu, tanh], regularization strengths [0.0001, 0.001, 0.01], and learning rates [0.001, 0.01]. The search evaluates 5 random configurations using 3-fold cross-validation with macro F1 scoring.

IV. Results

Model	Acc.	Prec.	Recall	F1
RF Baseline	0.785	0.62	0.48	0.54
Balanced LR	0.731	0.50	0.79	0.61
MLP NN	0.752	0.53	0.52	0.53

TABLE I. Churn class (class 1) performance comparison across models.

Table I summarizes the churn-class performance metrics. The baseline Random Forest achieves the highest overall accuracy (78.5%) but the lowest churn recall (48%). The class-weighted Logistic Regression achieves 79% recall, a 32-percentage-point improvement, at the cost of a 5.3-point accuracy reduction.

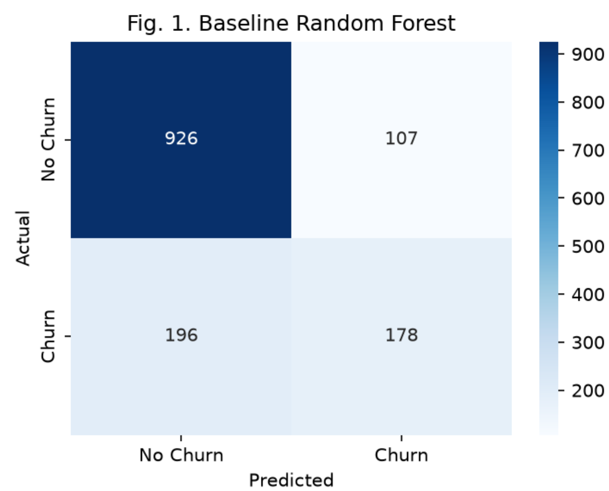

Fig. 1. Confusion matrix for the baseline Random Forest classifier.

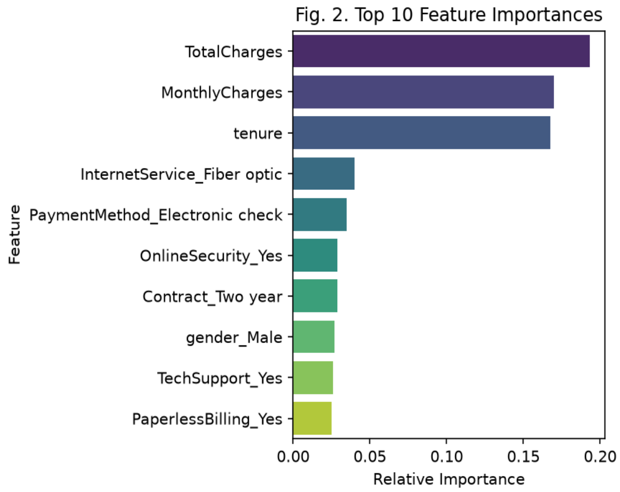

Fig. 2. Top 10 feature importances extracted from the Random Forest model.

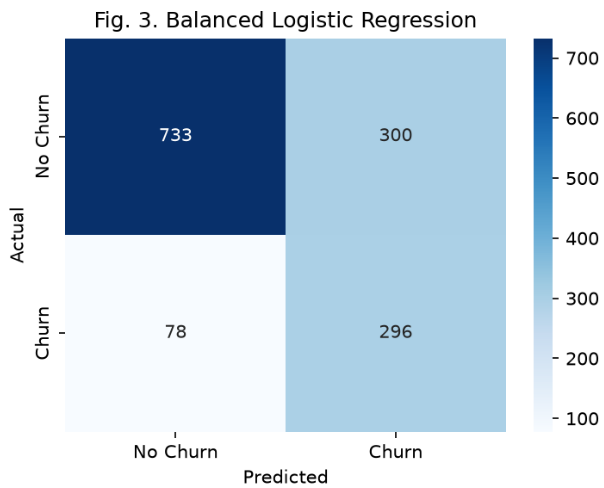

Fig. 3. Confusion matrix for the class-weighted Logistic Regression model.

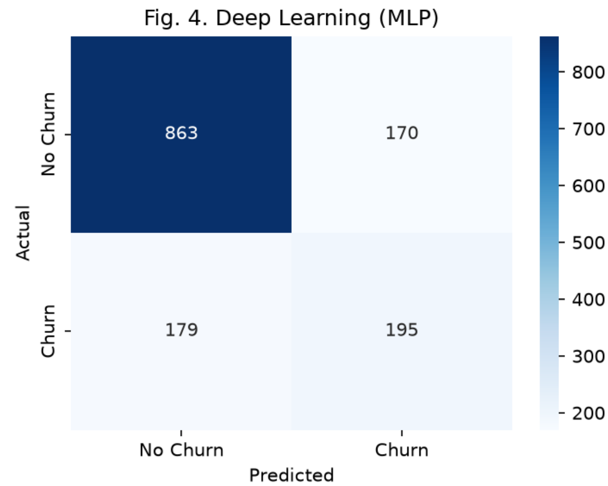

Fig. 4. Confusion matrix for the MLP neural network.

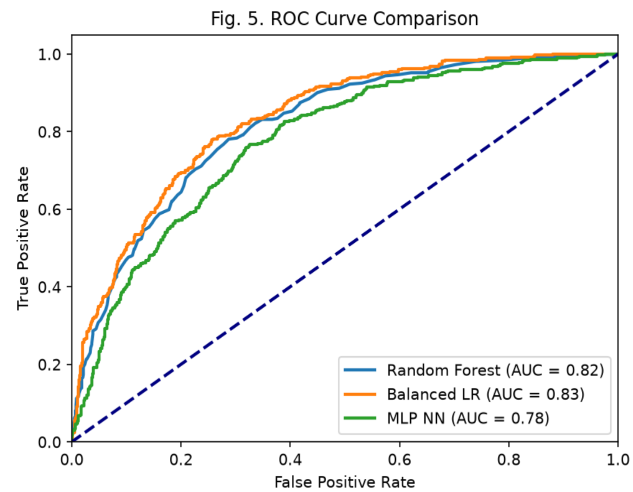

Fig. 5. ROC curves comparing all three models. AUC values are shown in the legend.

V. Discussion

The results confirm that overall accuracy is a misleading metric for imbalanced classification. The baseline Random Forest achieves 78.5% accuracy yet misses 52% of actual churners. In a business context where customer lifetime value averages \$780 and retention campaigns cost \$50 per customer, the additional churners identified by the Logistic Regression model yield a projected incremental profit of approximately \$83,950 per 1407 customers evaluated.

Feature importance analysis (Fig. 2) reveals that contract type, tenure, and monthly charges are the strongest churn predictors. Customers on month-to-month contracts with low tenure and high monthly charges represent the highest-risk segment. These insights directly inform targeted retention campaign design.

The MLP neural network underperforms expectations with 52% churn recall. This is likely attributable to the limited hyperparameter search budget (5 iterations) and the absence of explicit class balancing in the loss function. Future work should explore focal loss or SMOTE-augmented training to address this gap.

VI. Conclusion

This study demonstrates that a class-weighted Logistic Regression model provides the optimal balance of recall (79%) and interpretability for customer churn prediction on imbalanced telecommunications data. We recommend its immediate deployment as a replacement for accuracy-optimized baselines. Future work should investigate ensemble approaches combining the interpretability of linear models with the capacity of deep architectures, and validate findings on longitudinal data.

References

[1] A. Keramati, R. Jafari-Marandi, M. Aliannejadi, I. Ahmadian, M. Mozaffari, and U. Abbasi, "Improved churn prediction in telecommunication industry using data mining techniques," *Applied Soft Computing*, vol. 24, pp. 994-1012, 2014.

[2] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, 2002.

[3] IBM, "Telco Customer Churn," Kaggle, 2018. [Online]. Available: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

[4] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.